

Ivan Kirischian

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Website: bigbirdy7.github.io/myprojects

About me:

I am a high school student with a passion for robotics, coding, and technology, actively seeking opportunities to teach and mentor younger students. I am always eager to share my experience in designing, building, and programming robots and electronics projects. As a strong communicator and adaptable team player, my greatest soft skill is efficiently explaining complex concepts from the foundation up. To apply this, I run a solo tutoring business teaching math and programming to kids, while also regularly tutoring my peers. I also enjoy volunteering; recently, I ran the junior VEX V5 section for my school's STEM day, teaching visiting students how to drive the robots.

Skills

- Robotics/Electronics: VEX V5, Arduino, Mbot, Microbit, LEGO Mindstorms
- Programming: Java, Python, C++ and Block-based Coding (E.g. Scratch)
- Experienced with CAD tools like Tinkercad, SketchUp and Fusion360 (for VEX V5)
- Strong communication skills and patience when teaching others. My greatest soft skill is efficiently explaining complex concepts, starting from the foundations.
- I enjoy teaching, and always try to volunteer at positions that allow me to do so.

Job/Volunteering

- I run a solo tutoring business, teaching kids math and programming, while also regularly tutoring my sister and classmates.
- I always volunteer at any STEM related events, such as running the junior VEX V5 section for STEM day teaching the kids how the robot works and how to drive it, and open house for my Computer Engineering class, showcasing my projects.

Awards

Skills Regional Vex Competition– *[March, 2026]*

- Got Second at the YRSC for VEX V5 robotics, moving onto provincial competition.
- Led the junior team to create a functioning robot in less than 1 week for the school's STEM day.

McMaster Sumo Bot Competition: **First Place**– *[January, 2025]*

- Main developer for a robot controller to compete in an autonomous sumo wrestling competition.
- Collaborated with teammates to: build, debug, and optimize the robot operations.

School Awards

- Top of the class for Grade 12 Computer Science course (done in grade 11)
- Top of the class for Grade 11 Computer Science course (done in grade 10)

Personal Projects (bigbirdy7.github.io/myprojects)

- **3D Renderer and Flight Sim, From Scratch**
Programmed a 3D renderer in Java, completely from scratch, without copying tutorials. The only libraries it uses are a library for coloring pixels on a window. I learned about linear algebra, rendering and rasterising, then used my knowledge to create the project. Later I added multiplayer functionality.
- **Robot Car with Laser Turret**
Built a mobile robot with a laser turret controllable by a smartphone app, took apart servos and replaced inner components to achieve our project goal in a limited timeframe.
- **Real Life Portal Turret with AI vision (In-progress)**
A 3D printed turret that tracks a person as they walk through the room and “shoots” at them with LED lights. I greatly improved my soldering skills when making the project..
- **Math Quiz Device for Kids**
Created an educational game using an Arduino, keypad, and LCD to help my younger sister practice math, complete with a point system and a custom 3D-printed case.
- **Personal AI Assistant**
Desktop AI assistant using Groq API, microphone input, and a text-to-speech model. Inspired by the character GLaDOS from Portal 2. I am planning to integrate it into the Real Life Portal Turret.

Education

Thornhill Secondary School – Grade 11 Student

Graduation: 2027

- Relevant Courses: Computer Science 98% (grade 12) Computer Science 97% (grade 11), Computer Engineering 97% (grade 11) Computer Engineering 100% (grade 10)
- Member of Robotics Club (Will be president next year) and Coding Club